

Pankaj Thakur

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PROFESSIONAL SUMMARY

AI & Full-Stack Developer pursuing B.Tech in AI & Data Science at MITS Gwalior (2027). Specialises in **Generative AI**, **RAG**, and **multi-agent LLM systems** using **LangChain**, **LangGraph**, and **Groq**. Shipped 3 production-grade projects: a 4-module AI research platform (21 API routes, SSE streaming), a meeting intelligence platform (Whisper + LangGraph + RAG), and a full-stack parking SaaS (3-role RBAC, Razorpay, Socket.io). **Top 7.3% nationally in GATE CSE 2026**. Seeking **AI Engineer** / **SDE Internship**.

EDUCATION

Madhav Institute of Technology & Science (MITS), Gwalior

September 2023 – August 2027

B.Tech in Artificial Intelligence & Data Science

CGPA: 8.3 / 10

Relevant Coursework: DSA, DBMS, OOP, Operating Systems, Machine Learning, NLP, Computer Networks

TECHNICAL SKILLS

Languages	Python, C++ , JavaScript (ES6+), SQL, HTML5, CSS3
Frontend	React.js, Vite, Tailwind CSS, React Router DOM, Streamlit, Leaflet
Backend	Node.js, Express.js, FastAPI, REST API Design, JWT Authentication, RBAC, Zod
Databases	MongoDB Atlas, SQLite, MySQL, ChromaDB (Vector Store)
AI / GenAI	LangChain, LangGraph, RAG, HuggingFace Embeddings, Groq, OpenAI Whisper, Prompt Engineering, Agentic AI, Multi-Agent Systems, NLP
Tools	Git, GitHub, Docker, Postman, Razorpay, Google OAuth, Socket.io, SSE, Tavily API, FFmpeg
Deployment	Vercel, Render, MongoDB Atlas, Streamlit Cloud

PROJECTS

Summly — AI Meeting Intelligence Platform | [GitHub](#)

May 2026

FastAPI | LangGraph | Groq | OpenAI Whisper | ChromaDB | FFmpeg | SQLite | React.js | Streamlit | JWT

- Built **end-to-end meeting intelligence pipeline**; FFmpeg extraction (16 kHz WAV) → **spectral noise cleaning** (5–10% transcription accuracy gain) → **Whisper int8** transcription → **LangGraph + Groq** analysis generating **4 structured intelligence outputs** (summaries, action items, decisions, topic tags) per meeting.
- Implemented **RAG-powered Q&A** over **15+ indexed meeting transcripts** using 500-char chunking, **sentence-transformer embeddings**, and **ChromaDB top-3 semantic retrieval**, delivering Groq-grounded answers with **sub-3-second end-to-end response times**.
- Engineered **JWT + bcrypt multi-user auth**, SQLite schema (meetings/intelligence/users), **yt-dlp YouTube ingestion**, follow-up email generation, and **20+ REST endpoints** with structured error handling achieving **~80% automated test coverage**.

SmartPark — Full-Stack Parking SaaS | [GitHub](#)

April 2026

React.js | Node.js | Express.js | MongoDB Atlas | Socket.io | Razorpay | JWT | Zod | Leaflet | Vercel | Render

- Architected production parking SaaS with **3-role RBAC** (Driver / Owner / Admin); implemented **concurrency-safe booking** via **MongoDB ACID transactions + pessimistic locking** (`findOneAndUpdate` inside session), eliminating double-booking race conditions.
- Engineered **23 RESTful endpoints** (auth, parking, booking, payment, analytics, admin) with **JWT, Helmet, rate limiting, XSS/Mongo sanitisation**; **real-time availability** via **Socket.io broadcast** on every booking confirmation.
- Built **weighted geo-ranking engine** ($\text{dist} \times 0.5 + \text{price} \times 0.3 + \text{avail} \times 0.2$) powering **\$geoNear Smart Recommendations** across **8+ locations**; integrated **Razorpay HMAC-SHA256 webhook verification**, **Google OAuth**, and per-owner earnings analytics dashboard.

ResearchOS — Multi-Modal AI Research Platform | [GitHub](#)

March 2026

FastAPI | LangGraph | LangChain | Groq | React.js | ChromaDB | SQLite | SSE | JWT | Tavily API

- Architected **4-module AI research platform** (Topic Research, PDF Chat, News Intelligence, AI Dashboard) with **21 RESTful API routes**, JWT-secured SPA, and **real-time streaming design**; deployed on Render & Vercel.
- Engineered **async generator-ThreadPoolExecutor SSE bridge** sustaining **~500 tok/s Groq throughput** without event-loop blocking; **4-agent pipeline** (Search→Read→Write→Review) delivers cited Markdown reports with automated quality scoring.
- Implemented **7-stage RAG subsystem**: **HuggingFace all-MiniLM-L6-v2 (384-dim)** + **ChromaDB** per-UUID session isolation; custom **bind_tools()** loop bypassed hidden ReAct prompts causing Groq HTTP 400 errors.
- Integrated **Tavily News + Open-Meteo** into LangChain multi-tool Dashboard Agent with **API key rotation** and graceful fallback across **multiple live data sources**, achieving **zero-downtime LLM inference** throughout **100% of testing cycles**.

ACHIEVEMENTS & CERTIFICATIONS

- GATE CSE 2026** — Secured AIR 14,658 among 200,000+ candidates (**Top 7.3% nationally**), achieving a score of 426 with strong performance in Algorithms, DBMS, Operating Systems, and Computer Networks.
- LeetCode [Profile]** — Solved 120+ problems across arrays, trees, graphs, dynamic programming, hashing, and recursion with consistent Medium/Hard-level practice.
- DSA | PW Skills [Credential]** — Completed coursework covering data structures, algorithms, and time-space complexity analysis.
- Backend Development (Node.js & Express.js) | Udemy [Credential]** — Learned REST API design, server architecture, middleware patterns, authentication, authorization, and JWT-based security implementation.